THE PICARD GROUP OF A LOCALIZATION OF A REGULAR LOCAL RING IS TRIVIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted proof span refers to the class of L as '[L_f]' although L is already an R_f-module, and it uses the author's {\it free} italic markup in running text; both are delivered verbatim. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- A finitely presented module over a localization is the localization of a finite module over the base. invoked at proof lines 36787-36789. Produces the finite R-module M whose localization is L.
- A regular local ring has finite global dimension, so every finite module has a finite free resolution. invoked at proof lines 36790-36791. Turns M into a perfect complex, hence into a class in K_0(R_f).
- The K-group computation for perfect complexes and the determinant map on K_0. invoked at proof lines 36793-36797. Shows the class of L is a multiple of the class of R_f, and the determinant then forces triviality.

Source reference. Original lemma 05N5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7294–7294. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00O7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 885–885. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 36464: sources/stacks-more-algebra/pinned-original-source.tex, source lines 36464–36464. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 36491: sources/stacks-more-algebra/pinned-original-source.tex, source lines 36491–36491. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a regular local ring. Let $f \in R$. Then* $\mathrm{Pic}(R_f) = 0$.

Proof. Let L be an invertible R_f-module. In particular L is a finite R_f-module. There exists a finite R-module M such that $M_f \cong L$, see Algebra, Lemma 05N5. By Algebra, Proposition 00O7 we see that M has a finite free resolution $F_\bullet$ over R. It follows that L is quasi-isomorphic to a finite complex of *free* R_f-modules. Hence by Lemma 36491 we see that $[L_f] = n[R_f]$ in $K_0(R_f)$ for some $n \in \mathbf{Z}$. Applying the map of Lemma 36464 we see that L is trivial. $\square$